rl-matdesign — YAML Configuration Reference

Every config flag, what it does, and its choices

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Overview

A run is configured by one YAML file plus a few CLI flags. The three pluggable layers are chosen by **`env_type`** (the search space), **`predictor`** (the reward), and **`constraint_filter`** (action masking); each then reads its own keys. The RL and generation knobs are shared across scenarios.

- The YAML is the base; a few CLI flags override matching keys.
- **Seeds are CLI-only**: `--dp-seed` (predictor RNG), `--train-seed` (training; defaults to `--dp-seed`), `--gen-seed` (generation; defaults to `--dp-seed`).
- The full merged config is written to `<out>/run_config.json`.

CLI flags: `--config` (required), `--out` (required), `--method`, `--dp-seed`, `--train-seed`, `--gen-seed`, `--device` (cpu/cuda), `--dqn-loss`, `--dqn-augment-permutations`, `--save-checkpoint-freq`, `--resume-training`, `--only-generate`, `--skip-generation`, `--max-gen-attempts`, `--load-qnet` / `--load-policy` / `--load-value-net` / `--load-scaler`.

1. Environment

Key	What it does	Choices / default
`env_type`	Selects the environment	`fraction` (default, `CompositionEnv`); `integer_ratio` (`IntegerRatioEnv`); `multi_group` (`MultiGroupEnv`)

`fraction` env keys

Key	What it does	Choices / default
`cation_set`	Candidate element symbols	**required** (list)
`fraction_set`	Allowed fraction strings (grid)	default = built-in `0.05 … 0.80`
`total_units`	Grid resolution; step = `1/total_units`	`20` (->0.05); use `100` for 0.01
`n_components`	Distinct elements per episode	`5`
`anion_formula`	Fixed suffix appended to the formula	`""` (e.g. `O2H1`, `O3`)
`episode_style`	How each step picks	`element_then_amount` (default: choose element + amount); `fixed_order_amount` (order pinned to `cation_set`, choose only amounts)
`element_bounds`	Per-element `{el: [min,max]}` fraction caps	none; **only with `fixed_order_amount`**
`constraint_filter`	Action-masking filter (see §3)	`null`

`integer_ratio` env keys

Key	What it does	Choices / default
`cation_set`	Candidate elements	**required**
`ratio_set`	Allowed integer-ratio digits	default `"0" … "9"`
`n_components`	Elements per episode	`5`
`constraint_filter`	Action-masking filter	`null`

No sum-to-1 constraint; the composition is normalized afterward.

2. `multi_group` env — `groups:`

`groups:` is an **ordered list** of sublattice **groups**, filled in order. A group is the atomic unit; a single `fraction/integer_ratio` env is the one-group case. The predictor receives the structured `{group_name: {element: value}}`.

Common group keys

Key	What it does	Choices / default
`name`	Group label (appears in the structured terminal)	string
`kind`	Group type	`composition` (default) or `categorical`
`sites`	Sublattice size (atoms per formula unit) — bridges fraction<->count and assembles the chemical formula (`amount × sites`)	1
`constraint_filter` + its kwargs	Per-group filter (sees `prior_groups` for cross-group coupling)	(see §3)

`kind: composition` — pick N elements with amounts summing to 1. Reuses the `fraction`-env keys (`cation_set`, `fraction_set`, `total_units`, `n_components`, `episode_style`, `element_bounds`) plus friendly knobs:

Key	What it does	Choices / default
`amount`	Generate the value grid instead of writing `fraction_set`	`{min, max, step}` or a list
`host`	A host element auto-takes the leftover (list only the dopants; wires a `host_complement` filter)	element symbol

`kind: categorical` — pick discrete **real values**; no sum-to-1. Each slot is one element with its own value list; the terminal returns the chosen values unchanged (so a builder reads `Cl = 1.0` / `O = 1` directly).

Key	What it does
`choices`	`[{element, values: [...]}, …]` — one slot per element

A categorical filter may mask a slot by an earlier group (e.g. `sse_doping` masks the O slot by the P-site metal's category).

3. Constraint filters (`constraint_filter:`)

Value	What it does	Its keys
`null`	No constraint (default)	—
`smact_charge`	Keep only compositions that can be charge-neutral (SMACT oxidation states)	`smact_anions: [{symbol, charge, stoich}]` (or `smact_anion` / `smact_anion_charge` / `smact_anion_stoich`)

Value	What it does	Its keys
`last_step_element`	Force the last step to a required element	`required_elements`; `nonzero_ratio_at_last` (true); `reserve_for_last` (true)
`phase_pattern`	Keep compositions matching phase patterns	`phase_patterns`
`ooh_phase`	OOH oxyhydroxide phase screen	`target_phases` (default `[any]`)
`chain`	Apply several filters in sequence	`filters: [ {constraint_filter: …, …}, … ]`
`host_complement`	Dopants at a level, host takes the rest (wired automatically by a composition group's `host` knob)	`host_element`, `levels`
`sse_doping`	Mask the LiPS S-site O-form slot by the metal's category	`o_element` (0), `host_P` (P), `metal_only`, `oxide_only`
`"pkg.mod:Class"`	Your own filter (FQN)	your class's keys

4. Predictors (`predictor:`)

Value	What it does
`dummy`	Random reward (default; no models — smoke tests)
`structure_score`	**The one structure-based predictor** — build -> [relax] -> score N properties -> combine (see below)
`ooh`	OOH overpotential (adsorbate slabs)
`sinter_calcine`	RandomForest sintering/calcine temperature
`"pkg.mod:Class"`	Your own predictor (FQN)

`structure_score` keys

One predictor steered by dials. The pipeline is `builder.build -> [relax once, optional] -> score each property -> combine`. It replaces the former `dp_structure` / `dp_property` / `composite` / `structure_pipeline` (and the `hea` / `perovskite` aliases): each is now just a particular setting of the dials below.

Key	What it does	Default
`builder`	Builder name/FQN (`substitute` = fixed-lattice element swap; `sse` = doped supercell; or `pkg.mod:Class`)	`substitute`
`share_structure`	`true` = build+relax **one** cell, score all properties on it; `false` = each property builds its **own**	`true`
`n_random_configs`	Random placements per candidate	1
`geo_opt`	Relaxation stage (see below); omit / `enabled: false` to skip	optional
`properties`	Non-empty list of objective specs (see below)	required
`k`	Uncertainty coefficient	`1.0`
`base_poscar` / `site_symbol`	Builder knobs (for `substitute`); inherited by per-objective builders when `share_structure: false`	—

`geo_opt:` sub-keys — `model` (default `models/DPA-3.1-3M.pt`), `head` (user-defined), `fmax` (`0.001`), `steps` (`1000`), `relax_cell` (`true`), `enabled` (`true`).

`properties[]:` sub-keys:

Key	What it does	Default
`name`	Unique label (CSV column prefix)	`prop{i}`
`backend`	`energy` (DP potential energy via ASE) or `property` (DP `DeepProperty` head)	`property`
`models` (or legacy `dp_models`)	Ensemble checkpoint list	required
`head`	DP head (energy: calculator head; property: `DeepProperty` head)	none
`direction`	`max` / `min` (energy: use `min` for "lower is better")	`max`
`weight` / `scale`	Combine weight / unit scale	`1.0` / `1.0`
`objective`	`mean_minus_kstd` / `mean` / `mean_plus_kstd`	`mean_minus_kstd`

Key	What it does	Default
`energy_per_atom` *(energy backend)*	Normalize by atom count	`true`
`output_index` / `output_aggregator` *(property backend)*	Which vector component / collapse mode	`0` / `index` (`index`/`mean`/`max`)

Reward = `sum weight * objective_from_mean_std(direction*mean, std, objective, k) / scale` over the properties (identical formula in both `share_structure` regimes).

Migration cheatsheet (old -> `structure_score`):

Old predictor	Now
`dp_structure`	`builder: substitute`, one property `backend: energy`, `direction: min`
`dp_property`	`builder: substitute`, one property `backend: property`
`hea` / `perovskite`	as `dp_structure`, with `site_symbol: X` / `Fe`
`structure_pipeline`	`share_structure: true` (default), N `backend: property` objectives
`composite`	`share_structure: false`, the `objectives:` list becomes `properties:`

`ooh` keys

`base_poscar`, `dp_models`, `objective`, `k`, `n_random_configs`, `ads_height`, `ads_dz`, `geo_opt` (bool), `geo_opt_model`, `uncertainty` (`models`/`configs`/`total`), `output_index`, `target_phases`.

`sinter_calcine` keys

`rf_model` (joblib path), `mode` (`sinter` / `calcine`).

5. Builders (`builder:`)

A builder turns the agent's pick (a flat `{element: fraction}` or a structured `{group: {…}}`) into ASE structures. Built-ins:

`builder:`	What it does
`substitute`	Fixed-lattice element swap onto `site_symbol` placeholder sites (no vacancies). Keys: `base_poscar` (or `poscar`), `site_symbol` (default X).
`sse`	Doped-supercell recipe (P->metal, S->O/Cl/Br, charge-neutral Li vacancies). Keys below.
`"pkg.mod:Class"`	Your own builder (FQN) with `build(candidate, *, n_configs, rng)`.

`sse` builder keys

Key	What it does	Default
`base_poscar`	Base supercell	required
`valences`	Charge table `{el: int | {sulfide, oxide}}` (drives the Li-vacancy solve **and** oxide O count)	required
`halide_total`	Cl + Br per formula unit	`1.7`
`eligible_region`	Substitutable S region	`{symbol: S, take: last, count: 1000}`
`formula_units`	Formula units in the supercell — **inferred from the POSCAR** (host-P count / `p_site_per_fu`) unless set	inferred
`p_site_group` / `s_site_group`	Group names to read	`P_site` / `S_site`
`host`	`{P, S, Li}` host symbols	`{P:P, S:S, Li:Li}`
`p_site_per_fu` / `s_site_per_fu` / `li_per_fu`	Sites per formula unit	1 / 6 / 6

The S-site reads the categorical values directly — O is a form flag (`0` metal / `>0` oxide), Cl is a real per-f.u. count. `Br = halide_total − Cl` and the charge-neutral Li vacancy are derived. (No `cl_map` / `o_off`.)

6. RL method & hyperparameters

Key	What it does	Choices / default
`method`	Algorithm (CLI `--method` overrides)	`a2c` (default), `reinforce`, `dqn`

Policy gradient (`a2c` / `reinforce`)

Key	Default	Key	Default
`pg_warmup_eps`	200	`pg_entropy_coef`	0.01
`pg_num_iters`	1000	`pg_repeat_penalty_coef`	0.0
`pg_batch_eps`	15	`pg_repeat_penalty_shape`	`log` (also `sqrt`)
`pg_lr_actor`	0.001	`gamma` (or `pg_gamma`)	0.9
`pg_lr_critic`	0.001 (a2c only)		

DQN

Key	Default	Key	Default
dqn_lr	0.001	dqn_target_update_freq	100
dqn_hidden_dim	256	dqn_eps_anneal_eps	10000
dqn_batch_size	256	dqn_eps_min	0.05
dqn_warmup_eps	->pg_warmup_eps/500	dqn_gamma (or gamma)	0.9
dqn_num_train_eps	20000	dqn_loss	smoothl1 (also mse)
dqn_buffer_size	50000	dqn_augment_permutations	0
dqn_grad_steps_per_ep	5		

Generation (all methods)

Key	What it does	Default
num_gen_eps	Final candidates to generate	200
exploration_gen_eps	Extra high-temperature candidates	0
gen_temperature	Sampling temperature (DQN Boltzmann / PG)	1.0
gen_top_frac	Restrict sampling to top fraction of actions	0.0
gen_epsilon	ε-random during generation	0.0
k	Uncertainty coefficient for reward folding	1.0